

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Pipelines Sizing Check & Calculation Note

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Revision History

Rev No.	Date	Description of Revision
R01	21-07-22	First Issue for Client's Review
R02	20-12-22	Re-Issued for Client's Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-007

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-011	Steady State Simulation - Pipesim
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-013	Steady State Hydraulic Analysis - Hysys

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Abbreviations

Abbreviation	Meaning
API	American Petroleum Institute
DEP	Design and Engineering Practices
DP	Design Pressure
FEED	Front End Engineering
ISO	International Organisation for Standardization
MMscfd	Million standard cubic feet per day
NPS	Nominal Pipe Size
OD	Outer Diameter
TD	Design Temperature
FTHP	Flowing Tubing Head Pressure

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

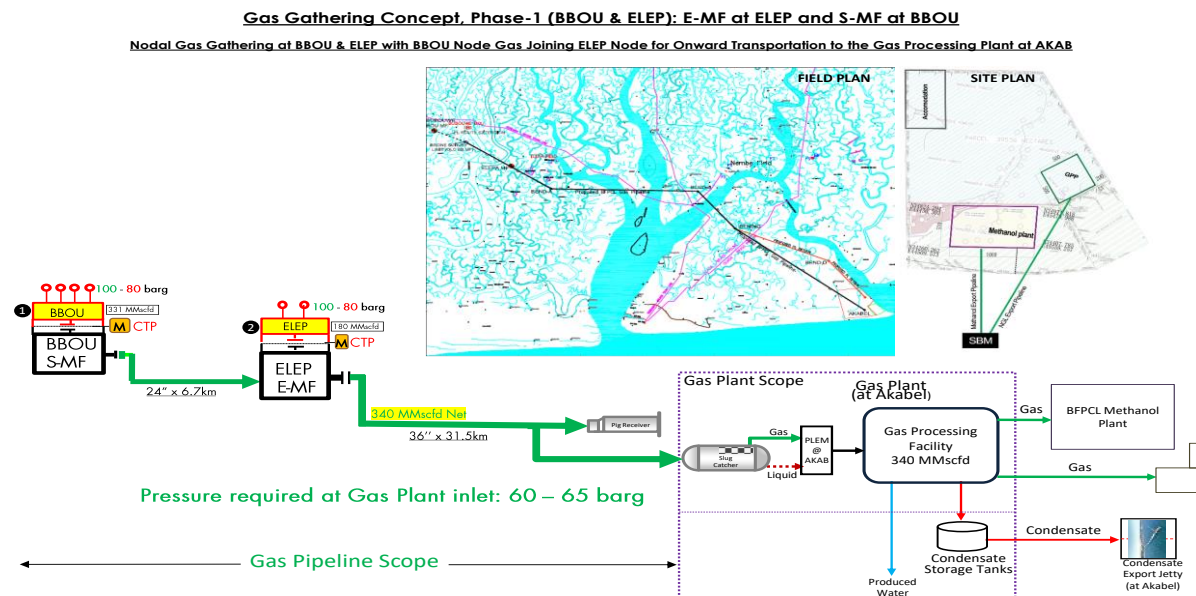


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of the document is to verify pipeline size that would deliver the gas to the Gas Plant at the required pressure of 60 – 65 barg.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP. 31.40.20.37-Gen	Line pipe for Critical Service (February 2019)
DEP. 82.00.10.10-Gen	Project Quality Assurance
DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline (February 2017)
ASME B31.8	Gas Transmission and Distribution Piping Systems
API 5L	Specification for Line Pipe
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.
ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Design Units of Measurement

Variables	Units
Barg	Bar Gauge (Pressure)
Bara	Bar Absolute (Pressure)
bbl	Barrel (Oil Volume)
BCF	Billion Cubic Feet
Bopd	Barrels of Oil per day
Bpd or B/D	Barrel per day
BS&W	Base Sediment and Water
cP	Centipoise (Unit of Viscosity)
F	Fahrenheit
C	Centigrade
km	kilometre
m	meter
Mbopd	Thousand Barrels of Oil per Day
Mbpd	Thousand Barrels of per Day
Mbwpd	Thousand Barrels of Water per Day
MJ	Megajoule
mm	Millimetre
MMscfd	Million Standard Cubic Feet Per Day
MW	Mega-Watt
ppm	Part Per Million
psig	Pounds Per Square Inch Gauge (Pressure)
Component	Vol%, mol%, wt ppm
Liquid flow	Kg/h, m ³ /h
Steam flow	Kg/h
Level	%, m, mm
Density	Kg/m ³
Conductivity	μs/cm
Weight	Kg, t
Distance (length)	m, mm
Power	W, KW
Force	N, kN
Current	A, mA
Energy	MJ, kJ
Sound	dBA
Frequency	HZ

3. The Pipeline Sizing

3.1. Objective

The objective in this case is to select the line size that will moderate pressure drop in a way that the gas will arrive its destination at the range of pressures required.

The general principle governing this is the variety of industry standard Steady State Flow Equations, tailored to suit prevailing flow parameters and conditions, and the pipeline size range.

3.2. Steady State Flow Equation Decision Matrix

Three varieties of these equations are widely used in natural gas transmission:

- i) Panhandle A - used for medium to long pipeline length
- ii) Panhandle B - used for very long and distant pipeline length
- iii) Weymouth - used for short distribution systems and pipings. It is inaccurate for long distance pipelines, as it tends to overestimate pressure drop in long pipelines.

Each equation has their various applicability criteria for pressure drop calculations. The applicability criteria are as follows;

Panhandle A

- i) Diameter size range of 15 – 24 inches (381 – 609.6mm)
- ii) Pressure between 800 – 1500 psi (55 – 103 bar)
- iii) Reynolds number ranges from 4,000 – 40,000,000

Panhandle B

- i) Diameter size is greater than 24 inches (609.6mm)
- ii) Pressure is greater than 1000 psi (69 bar)
- iii) Reynolds number ranges from 40,000 – 40,000,000

Weymouth

- i) Diameter size is less than 15 inches (381mm)
- ii) Pressure is less than 800 psi (55 bar)
- iii) Reynolds number is less than 4,000,000

3.2.1. Equation Selection Decision Matrix: Case-1 FTHP 100 barg

100 barg FTHP		PANHANDLE A			PANHANDLE B			WEYMOUTH		
		Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number
24" x 6.7km BBOU-ELEP Pipeline Segment	Diameter (in)	15 - 24	800 - 1,500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	24	✓			✗			✗		
	Pressure (psia)									
	1,465		✓			✓			✗	
	Reynolds Number			✓			✓			✗
	16,898,880									
36" x 31.5km ELEP-AKAB Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	36	✗			✓			✗		
	Pressure (psia)									
	1,465		✓			✓			✗	
	Reynolds Number			✓			✓			✗
	13,139,824									
32" x 31.5km ELEP-AKAB Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	32	✗			✓			✗		
	Pressure (psia)									
	1,465		✓			✓			✗	
	Reynolds Number			✓			✓			✗
	14,664,216									
28" x 31.5km ELEP-AKAB Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	28	✗			✓			✗		
	Pressure (psia)									
	1,465		✓			✓			✗	
	Reynolds Number			✓			✓			✗
	16,793,600									
24" x 31.5km ELEP-AKAB Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	24	✓			✗			✗		
	Pressure (psia)									
	1,465		✓			✓			✗	
	Reynolds Number			✓			✓			✗
	19,706,851									

3.2.2. Equation Selection Decision Matrix: Case-2 FTHP 80 barg

80 barg FTHP		PANHANDLE A			PANHANDLE B			WEYMOUTH		
		Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number
24" x 6.7km BBOU-ELEP Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	24	✓			✗			✗		
	Pressure (psia)		✓			✓			✗	
	1,175									
	Reynolds Number			✓			✓			✗
	16,898,880									
		PANHANDLE A			PANHANDLE B			WEYMOUTH		
		Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number
36" x 31.5km ELEP-AKAB Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	36	✗			✓			✗		
	Pressure (psia)		✓			✓			✗	
	1,175									
	Reynolds Number			✓			✓			✗
	13,139,824									
		PANHANDLE A			PANHANDLE B			WEYMOUTH		
		Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number
32" x 31.5km ELEP-AKAB Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	32	✗			✓			✗		
	Pressure (psia)		✓			✓			✗	
	1,175									
	Reynolds Number			✓			✓			✗
	14,664,216									
		PANHANDLE A			PANHANDLE B			WEYMOUTH		
		Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number
28" x 31.5km ELEP-AKAB Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	28	✗			✓			✗		
	Pressure (psia)		✓			✓			✗	
	1,175									
	Reynolds Number			✓			✓			✗
	16,793,600									
		PANHANDLE A			PANHANDLE B			WEYMOUTH		
		Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number	Diameter (in)	Pressure (psia)	Reynolds Number
28" x 31.5km ELEP-AKAB Pipeline Segment	Diameter (in)	15 - 24	800 - 1500	4,000 - 40,000,000	> 24	> 1000	40,000 - 40,000,000	< 15	< 800	< 4,000,000
	24	✓			✗			✗		
	Pressure (psia)		✓			✓			✗	
	1,175									
	Reynolds Number			✓			✓			✗
	19,706,851									

3.2.3. Summary of Equations Selection

Pipeline Segment	Pressure loss equation selected
24" x 6.7km BBOU – ELEPA	Panhandle A
36" x 31.5km ELEPA - AKABEL	Panhandle B
32" x 31.5km ELEPA - AKABEL	Panhandle B
28" x 31.5km ELEPA - AKABEL	Panhandle B
24" x 31.5km ELEPA - AKABEL	Panhandle A

3.3. Methodology

The pressure drops along pipeline is calculated in small incremental intervals, - 25 m intervals for the 24" x 4.2 km stretch of Brone surveyed BBOU-ELEP, and 100 m intervals for 36" x 31.5 km ELEP-AKAB, representing 160 and 302 segments respectively.

The reason for this method is simple, - to ensure representative averages of parameters of the equation, especially the gas compressibility factor which poses the greatest challenge to equation of state prediction.

Gas compressibility factor charts / graph and calculators were also used.

The Pressure Loss calculation was done both for the single phase gas flow, and for two-phase flow, representing the situation where liquid is produced with the gas.

Details of all calculations are in the excel workbook accompanying this report.

3.4. Summary of Results

The results of the pipe flow paremeters are summarized below:

Field Manifolds & End Manifold Pressures													
Case-1 Nodes THP of 100 barg				Single Phase Gas Flow					Two Phase Gas Flow				
				Pressures (barg)			Pressure Drop		Pressures (barg)			Pressure Drop	
		Segment	Pipe ID	BBOU M/F	ELEP M/F	AKAB	barg	barg / km	BBOU M/F	ELEP M/F	AKAB	barg	barg / km
		24" x 6.7 km BBOU - ELEP	22.77"	100.00	98.01		1.99	0.30	100.00	97.77		2.23	0.33
		24" x 6.7 km BBOU - ELEP with 36" x 31.5 km ELEP - AKAB	34.15"		98.01	88.63	9.38	0.30		97.77	88.08	9.69	0.31
		24" x 6.7 km BBOU - ELEP with 32" x 31.5 km ELEP - AKAB	30.60"		98.01	82.16	15.85	0.50		97.77	81.59	16.19	0.51
		24" x 6.7 km BBOU - ELEP with 28" x 31.5 km ELEP - AKAB	26.72"		98.01	63.62	34.39	1.09		97.77	63.01	34.77	1.10
		24" x 6.7 km BBOU - ELEP with 24" x 31.5 km ELEP - AKAB	22.77"		98.01	43.18	54.83	1.74		97.77	42.68	55.10	1.75
Pressure Required (barg)				60 - 65			60 - 65						
Case-2 Nodes THP of 80 barg				Single Phase Gas Flow					Two Phase Gas Flow				
				Pressures (barg)			Pressure Drop		Pressures (barg)			Pressure Drop	
		Segment	Pipe ID	BBOU M/F	ELEP M/F	AKAB	barg	barg / km	BBOU M/F	ELEP M/F	AKAB	barg	barg / km
		24" x 6.7 km BBOU - ELEP	22.77"	80.00	78.34		1.66	0.25	80.00	78.00		2.00	0.30
		24" x 6.7 km BBOU - ELEP with 36" x 31.5 km ELEP - AKAB	34.15"		78.34	68.91	9.42	0.30		78.00	68.17	9.83	0.31
		24" x 6.7 km BBOU - ELEP with 32" x 31.5 km ELEP - AKAB	30.60"		78.34	62.67	15.66	0.50		78.00	62.14	15.86	0.50
		24" x 6.7 km BBOU - ELEP with 28" x 31.5 km ELEP - AKAB	26.72"		78.34	42.36	35.98	1.14		78.00	41.73	36.27	1.15
		24" x 6.7 km BBOU - ELEP with 24" x 31.5 km ELEP - AKAB	22.77"		78.34	23.42	54.91	1.74		78.00	22.85	55.16	1.75
Legend													
Line Size Meets Requirement with reasonable margin													
Borderline Case with No Margin													
Does Not Meet Requirement													

4. Pressure Calculation Basis (Single phase Flow)

4.1. Pressure Calculation Basis for BBOU – ELEPA Pipeline Segment:

A line size of 24" is selected for the BBOU - ELEPA Pipeline Segment.

4.2. Efficiency Factor (E)

An efficiency of factor of 0.95 (95%) is used as the pipeline is in very good operating condition.

4.3. Inside Diameter

An inside diameter of 22.77 inches (578.36 mm) shall be used as per Pipeline Wall Thickness Calculations.

4.4. Gas specific Gravity

A gas specific gravity of 0.62 shall be used as per PVT data from Production Forecast and Fluid Compositions analysis.

4.5. Reynolds Number (Re)

The Reynolds Number is calculated using this formula;

$$\mathbf{Re} = 45 \times \frac{Q_b \times G}{D}$$

Where,

Q_b - Flow rate at standard conditions (ft³/hr)

G - Gas specific gravity, dimensionless

D – Inside Diameter (inches)

4.6. Gas Compressibility (Z) factor

The gas compressibility (Z) factor is calculated based on Dranchuk and Abou-Kassem equation of state (DAK - EOS). It is calculated at every 25 metre interval.

4.7. 24" BBOU - ELEPA Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data are as shown below

Parameter	Description
NPS	24"
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.

Flow rate	331,000,000 SCFD (331 MMscfd)
Design Pressure	110 barg
Inlet Pressure	100 barg
Base Pressure	1.79 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	22.77 inches
Efficiency Factor	0.95

4.8. 24" BBOU - ELEPA Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below

Parameter	Description
NPS	24"
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	331,000,000 SCFD (331 MMscfd)
Design Pressure	110 barg
Inlet Pressure	80 barg
Base Pressure	1.45 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	22.77 inches
Efficiency Factor	0.95

4.9. Pipeline Material

The line pipe is carbon steel material grade of API-5L X70 with Product Specification Level - PSL 2 with a corrosion allowance of 3mm. The pipeline is designed for a pressure of 110 barg and the design temperature shall be 60°C.

4.10. Pressure Calculation Basis for ELEPA – AKABEL Pipeline Segment:

Line sizes of 36", 32", 28" and 24" are selected for the ELEPA – AKABEL Pipeline Segment.

4.11. Efficiency Factor (E)

An efficiency of factor of 0.95 (95%) is used as the pipeline is in very good operating condition.

4.12. Inside Diameter

The following inside diameter sizes for each line size shall be used as per Pipeline Wall Thickness Calculations.

S/n	Line Size	Inside Diameter
1	36"	34.15"
2	32"	30.60"
3	28"	26.72"
4	24"	22.77"

4.13. Gas specific Gravity

A gas specific gravity of 0.62 shall be used as per PVT data from Production Forecast and Fluid Compositions analysis.

4.14. Reynolds Number (Re)

The Reynolds Number is calculated using this formula;

$$\mathbf{Re} = 45 \times \frac{Q_b \times G}{D}$$

Where,

Q_b - Flow rate at standard conditions (ft³/hr)

G - Gas specific gravity, dimensionless

D – Inside Diameter (inches)

4.15. Gas Compressibility (Z) factor

The gas compressibility (Z) factor is calculated based on Dranchuk and Abou-Kassem equation of state (DAK - EOS). It is calculated at every 100 metre interval.

4.16. 36" ELEPA - AKABEL Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
NPS	36"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 SCFD (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	98.01 barg
Base Pressure	3.96 barg
Design Temperature	60°C
Base Temperature	19.07°C
Gas specific gravity	0.62
Pipeline Inside Diameter	34.15 inches
Efficiency Factor	0.95

4.17. 36" ELEPA - AKABEL Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below

Parameter	Description
NPS	36"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 SCFD (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	78.34 barg

Base Pressure	3.62 barg
Design Temperature	60°C
Base Temperature	26.85°C
Gas specific gravity	0.62
Pipeline Inside Diameter	34.15 inches
Efficiency Factor	0.95

4.18. 32" ELEPA - AKABEL Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	32"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 SCFD (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	98.01 barg
Base Pressure	3.96 barg
Design Temperature	60°C
Base Temperature	19.07°C
Gas specific gravity	0.62
Pipeline Inside Diameter	30.60 inches
Efficiency Factor	0.95

4.19. 32" ELEPA - AKABEL Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below

Parameter	Description
NPS	32"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 SCFD (386 MMscfd)

Design Pressure	110 barg
Inlet Pressure	78.34 barg
Base Pressure	3.62 barg
Design Temperature	60°C
Base Temperature	26.85°C
Gas specific gravity	0.62
Pipeline Inside Diameter	30.60 inches
Efficiency Factor	0.95

4.20. 28" ELEPA - AKABEL Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	28"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 SCFD (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	98.01 barg
Base Pressure	3.96 barg
Design Temperature	60°C
Base Temperature	19.07°C
Gas specific gravity	0.62
Pipeline Inside Diameter	26.72 inches
Efficiency Factor	0.95

4.21. 28" ELEPA - AKABEL Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	28"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km

Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 SCFD (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	78.34 barg
Base Pressure	3.62 barg
Design Temperature	60°C
Base Temperature	26.85°C
Gas specific gravity	0.62
Pipeline Inside Diameter	26.72 inches
Efficiency Factor	0.95

4.22. 24" ELEPA - AKABEL Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	24"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 SCFD (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	98.01 barg
Base Pressure	3.10 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	26.72 inches
Efficiency Factor	0.95

4.23. 24" ELEPA - AKABEL Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	24"
From	ELEPA Manifold Station

To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 SCFD (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	78.34 barg
Base Pressure	2.66 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	26.72 inches
Efficiency Factor	0.95

4.24. Pipeline Material

The line pipe is carbon steel material grade of API-5L X70 with Product Specification Level - PSL 2 with a corrosion allowance of 3mm. The pipeline is designed for a pressure of 110 barg and the design temperature shall be 60°C.

5. Pressure Drop Calculation (Single phase Flow)

5.1. Pressure Drop calculation procedure (BBOU – ELEPA Pipeline Segment)

For determination of the required pressure drop at the BBOU – ELEPA Pipeline Segment the following pressure loss equation is used;

Panhandle A

It is used in pipelines with a diameter size range of 15 – 24 inch, pressure values of 800 – 1500 psi and Reynolds number ranging from 4,000 and 40,000,000.

The Panhandle A equation is defined as;

$$Q = 435.87 \times (T_b/P_b)^{1.0778} \times D^{2.6182} \times E \times \left[(P_1^2 - P_2^2 - (0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / (T_{avg} \times Z_{avg})) / (G^{0.8539} \times L \times T_{avg} \times Z_{avg}) \right]^{0.5394}$$

Parameters Definition and its Units

P_1 – Inlet Pressure (psia)

P_2 – Outlet Pressure (psia)

Q – Gas Flowrate (SCFD)

T_b – Base Temperature (Rankine)

P_b – Base Pressure (psia)

P_{avg} – Average Pressure (psia)

T_{avg} – Average Temperature (Rankine)

Z_{avg} – Average Gas Compressibility Factor

G – Gas specific gravity

h_2-h_1 – Elevation difference (ft)

L – Line Length (mile)

D – Inside Diameter (inch)

E – Efficiency

To find P_2 ,

Let the following parts of the equation be represented by the letters A, B and C

$$A = \left(0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2 \right) / T_{avg} \times Z_{avg}$$

$$B = G^{0.8539} \times L \times T_{avg} \times Z_{avg}$$

$$C = \left((T_b / P_b)^{1.0788} \right) \times D^{2.6182} \times E$$

Therefore;

$$Q = 435.87 \times C \times \left[(P_1^2 - P_2^2 - A) / B \right]^{0.5394}$$

Divide both sides by $435.87 \times C$

$$Q / (435.87 \times C) = \left[(P_1^2 - P_2^2 - A) / B \right]^{0.5394}$$

Take the $^{0.5394}\sqrt{}$ of both sides

$$\left[Q / (435.87 \times C) \right]^{1/0.5394} = \frac{P_1^2 - P_2^2 - A}{B}$$

$$\left[Q / (435.87 \times C) \right]^{1.853912} = \frac{P_1^2 - P_2^2 - A}{B}$$

Cross multiply

$$\left[Q / (435.87 \times C) \right]^{1.853912} \times B = P_1^2 - P_2^2 - A$$

$$P_2^2 = \left[P_1^2 - A - \left(Q / (435.87 \times C) \right)^{1.853912} \times B \right]$$

Therefore;

$$P_2 = \sqrt{P_1^2 - A - \left(\left(Q / (435.87 \times C) \right)^{1.853912} \times B \right)}$$

$$P_2 = \sqrt{P_1^2 - \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right) - \left[\left(Q / (435.87 \times ((T_b/P_b)^{1.0788}) \times D^{2.6182} \times E) \right)^{1.853912} \times (G^{0.8539} \times L \times T_{avg} \times Z_{avg}) \right]}$$

Steps to calculate Outlet Pressure (P₂)

- 1) Convert the parameters to their field units;

Barg to psi (pressure)

Celsius to Rankine (temperature)

Metre to miles (length)

Metre to feet (Elevation difference)

- 2) Calculate the values of A, B and C;

$$A = \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right)$$

$$B = G^{0.8539} \times L \times T_{avg} \times Z_{avg}$$

$$C = ((T_b/P_b)^{1.0788}) \times D^{2.6182} \times E$$

- 3) Using the values of A, B and C, calculate the outlet pressure (P₂) in psi

$$P_2 = \sqrt{P_1^2 - A - \left(\left(Q / (435.87 \times C) \right)^{1.853912} \times B \right)}$$

- 4) Convert P₂ from psi to bar

5.2. Pressure Drop calculation procedure (ELEPA - AKABEL Pipeline Segment)

For determination of the required pressure drop in the ELEPA - AKABEL Pipeline Segment the Panhandle B is used with the exception of the 24" line size in which Panhandle A is used:

Panhandle B

It is used in pipelines with a diameter size greater than 24 inch, pressure values greater than 1000 psi and Reynolds number ranging from 40,000 and 40,000,000.

The Panhandle B equation is defined as;

$$Q = 737 \times (T_b/P_b)^{1.02} \times D^{2.53} \times E \times \left[(P_1^2 - P_2^2 - \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right)) / (G^{0.961} \times L \times T_{avg} \times Z_{avg}) \right]^{0.51}$$

Parameters Definition and its units

P_1 – Inlet Pressure (psia)

P_2 – Outlet Pressure (psia)

Q – Gas Flowrate (SCFD)

T_b – Base Temperature (Rankine)

P_b – Base Pressure (psia)

P_{avg} – Average Pressure (psia)

T_{avg} – Average Temperature (Rankine)

Z_{avg} – Average Gas Compressibility Factor

G – Gas specific gravity

h_2-h_1 – Elevation difference (ft)

L – Line Length (mile)

D – Inside Diameter (inch)

E – Efficiency

To find P_2 ,

Let the following parts of the equation be represented by the letters A, B and C

$$A = \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right)$$

$$B = G^{0.961} \times L \times T_{avg} \times Z_{avg}$$

$$C = \left((T_b/P_b)^{1.02} \right) \times D^{2.53} \times E$$

Therefore;

$$Q = 737 \times C \times \left[(P_1^2 - P_2^2 - A) / B \right]^{0.51}$$

Divide both sides by $737 \times C$

$$Q / (737 \times C) = [(P_1^2 - P_2^2 - A) / B]^{0.51}$$

Take the $^{0.51}\sqrt{}$ of both sides

$$[Q / (737 \times C)]^{1/0.51} = \frac{P_1^2 - P_2^2 - A}{B}$$

$$[Q / (737 \times C)]^{1.9608} = \frac{P_1^2 - P_2^2 - A}{B}$$

Cross multiply

$$[Q / (737 \times C)]^{1.9608} \times B = P_1^2 - P_2^2 - A$$

$$P_2^2 = [P_1^2 - A - ((Q / (737 \times C))^{1.9608} \times B)]$$

Therefore;

$$P_2 = \sqrt{[P_1^2 - A - ((Q / (737 \times C))^{1.9608} \times B)]}$$

$$P_2 = \sqrt{P_1^2 - ((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg}) - [((Q / (737 \times ((T_b/P_b)^{1.02}) \times D^{2.53} \times E))^{1.9608} \times (G^{0.961} \times L \times T_{avg} \times Z_{avg}))]}$$

Steps to calculate Outlet Pressure (P_2)

1) Convert the parameters to their field units;

Barg to psi (pressure)

Celsius to Rankine (temperature)

Metre to miles (length)

Metre to feet (Elevation)

2) Calculate the values of A, B and C;

$$A = \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right)$$

$$B = G^{0.961} \times L \times T_{avg} \times Z_{avg}$$

$$C = (T_b/P_b)^{1.02} \times D^{2.53} \times E$$

3) Using the values of A, B and C, calculate the outlet pressure (P₂) in psi

$$P_2 = \sqrt{P_1^2 - A - \left((Q / (737 \times C))^{1.9608} \times B \right)}$$

4) Convert P₂ from psi to bar

5.3. Summary and Recommendation

The pressure drop calculation is used to determine the pipeline sizing and based on these calculations, the recommended line sizes are 24 inch for BBOU – ELEPA Pipeline Segment with 28 inch and 32 inch for ELEPA – AKABEL Pipeline Segment at a FTHP of 100 barg and 80 barg respectively.

Pipeline size (inch) BBOU - ELEPA	Pipeline size (inch) ELEPA - AKABEL	Inlet Pressure (barg) at BBOU	Inlet Pressure (barg) at ELEPA	Delivery Pressure (barg) at AKABEL
24	28	100	98.01	63.62
24	32	80	78.34	62.67

6. Pressure Calculation Basis (Two phase Flow)

6.1. Pressure Calculation Basis for BBOU – ELEPA Pipeline Segment:

A line size of 24" is selected for the BBOU - ELEPA Pipeline Segment.

6.2. Efficiency Factor (E)

The efficiency factor for each interval of the pipeline is determined using the Flanigan method. The Superficial gas velocity (U) and liquid gas ratio (R) are calculated. From these two parameters, the Panhandle efficiency is gotten from the Flanigan Efficiency Factor chart/graph.

6.3. Inside Diameter

An inside diameter of 22.77 inches (578.36 mm) shall be used as per Pipeline Wall Thickness Calculations.

6.4. Gas specific Gravity

A gas specific gravity of 0.62 shall be used as per PVT data from Production Forecast and Fluid Compositions analysis.

6.5. Gas Compressibility (Z) factor

The gas compressibility (Z) factor is calculated based on Dranchuk and Abou-Kassem equation of state (DAK - EOS). It is calculated at every 25 metre interval.

6.6. 24" BBOU - ELEPA Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data are as shown below

Parameter	Description
NPS	24"
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	331,000,000 SCFD (331 MMscfd)
Design Pressure	110 barg
Inlet Pressure	100 barg
Base Pressure	1.45 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62

Pipeline Inside Diameter	22.77 inches
Efficiency Factor	0.95

6.7. 24" BBOU - ELEPA Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below

Parameter	Description
NPS	24"
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	331,000,000 scfd (331 MMscfd)
Design Pressure	110 barg
Inlet Pressure	80 barg
Base Pressure	1.28 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	22.77 inches
Efficiency Factor	0.95

6.8. Pipeline Material

The line pipe is carbon steel material grade of API-5L X70 with Product Specification Level - PSL 2 with a corrosion allowance of 3mm. The pipeline is designed for a pressure of 110 barg and the design temperature shall be 60°C.

6.9. Pressure Calculation Basis for ELEPA – AKABEL Pipeline Segment:

Line sizes of 36", 32", 28" and 24" are selected for the ELEPA – AKABEL Pipeline Segment.

6.10. Efficiency Factor (E)

The efficiency factor for each interval of the pipeline is determined using the Flanigan method. The Superficial gas velocity (U) and liquid gas ratio (R) are calculated. From these two parameters, the Panhandle efficiency is gotten from the Flanigan Efficiency Factor chart/graph.

6.11. Inside Diameter

The following inside diameter sizes for each line size shall be used as per Pipeline Wall Thickness Calculations.

S/n	Line Size	Inside Diameter
1	36"	34.15"
2	32"	30.60"
3	28"	26.72"
4	24"	22.77"

6.12. Gas specific Gravity

A gas specific gravity of 0.62 shall be used as per PVT data from Production Forecast and Fluid Compositions analysis.

6.13. Gas Compressibility (Z) factor

The gas compressibility (Z) factor is calculated based on Dranchuk and Abou-Kassem equation of state (DAK - EOS). It is calculated at every 100 metre interval.

6.14. 36" ELEPA - AKABEL Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
NPS	36"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 scfd (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	98.01 barg
Base Pressure	2.64 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	34.15 inches
Efficiency Factor	0.95

6.15. 36" ELEPA - AKABEL Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below

Parameter	Description
NPS	36"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 scfd (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	78.34 barg
Base Pressure	2.49 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	34.15 inches

Efficiency Factor	0.95
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6.16. 32" ELEPA - AKABEL Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	32"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 scfd (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	98.01 barg
Base Pressure	2.64 barg
Design Temperature	60°C
Base Temperature	19.07°C
Gas specific gravity	0.62
Pipeline Inside Diameter	30.60 inches
Efficiency Factor	0.95

6.17. 32" ELEPA - AKABEL Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below

Parameter	Description
NPS	32"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 scfd (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	78.34 barg
Base Pressure	2.49 barg
Design Temperature	60°C
Base Temperature	15.74°C

Gas specific gravity	0.62
Pipeline Inside Diameter	30.60 inches
Efficiency Factor	0.95

6.18. 28" ELEPA - AKABEL Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	28"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 scfd (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	98.01 barg
Base Pressure	2.64 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	26.72 inches
Efficiency Factor	0.95

6.19. 28" ELEPA - AKABEL Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	28"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 scfd (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	78.34 barg
Base Pressure	2.49 barg

Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	26.72 inches
Efficiency Factor	0.95

6.20. 24" ELEPA - AKABEL Pipeline Segment Data (FTHP of 100 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	24"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 scfd (386 MMscfd)
Design Pressure	110 barg
Inlet Pressure	78.34 barg
Base Pressure	2.49 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	26.72 inches
Efficiency Factor	0.95

6.21. 24" ELEPA - AKABEL Pipeline Segment Data (FTHP of 80 barg)

The pipeline design data are as shown below;

Parameter	Description
NPS	24"
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Flow rate	386,000,000 scfd (386 MMscfd)
Design Pressure	110 barg

Inlet Pressure	78.34 barg
Base Pressure	2.49 barg
Design Temperature	60°C
Base Temperature	15.74°C
Gas specific gravity	0.62
Pipeline Inside Diameter	26.72 inches
Efficiency Factor	0.95

6.22. Pipeline Material

The line pipe is carbon steel material grade of API-5L X70 with Product Specification Level - PSL 2 with a corrosion allowance of 3mm. The pipeline is designed for a pressure of 110 barg and the design temperature shall be 60°C.

7. Pressure Drop Calculation (Two phase Flow)

7.1. Pressure Drop calculation procedure (BBOU – ELEPA Pipeline Segment)

For determination of the required pressure drop at the BBOU – ELEPA Pipeline Segment the following pressure loss equation is used;

Panhandle A

It is used in pipelines with a diameter size range of 15 – 24 inch, pressure values of 800 – 1500 psi and Reynolds number ranging from 4,000 and 40,000,000.

The Panhandle A equation is defined as;

$$Q = 435.87 \times (T_b/P_b)^{1.0778} \times D^{2.6182} \times E \times \left[(P_1^2 - P_2^2 - (0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg}) / (G^{0.8539} \times L \times T_{avg} \times Z_{avg}) \right]^{0.5394}$$

Parameters Definition and its Units

P_1 – Inlet Pressure (psia)

P_2 – Outlet Pressure (psia)

Q – Gas Flowrate (SCFD)

T_b – Base Temperature (Rankine)

P_b – Base Pressure (psia)

P_{avg} – Average Pressure (psia)

T_{avg} – Average Temperature (Rankine)

Z_{avg} – Average Gas Compressibility Factor

G – Gas specific gravity

h_2-h_1 – Elevation difference (ft)

L – Line Length (mile)

D – Inside Diameter (inch)

E – Efficiency

To find P_2 ,

Let the following parts of the equation be represented by the letters A, B and C

$$A = \left(0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2 \right) / T_{avg} \times Z_{avg}$$

$$B = G^{0.8539} \times L \times T_{avg} \times Z_{avg}$$

$$C = \left((T_b / P_b)^{1.0788} \right) \times D^{2.6182} \times E$$

Therefore;

$$Q = 435.87 \times C \times \left[(P_1^2 - P_2^2 - A) / B \right]^{0.5394}$$

Divide both sides by $435.87 \times C$

$$Q / (435.87 \times C) = \left[(P_1^2 - P_2^2 - A) / B \right]^{0.5394}$$

Take the $^{0.5394}\sqrt{}$ of both sides

$$\left[Q / (435.87 \times C) \right]^{1/0.5394} = \frac{P_1^2 - P_2^2 - A}{B}$$

$$\left[Q / (435.87 \times C) \right]^{1.853912} = \frac{P_1^2 - P_2^2 - A}{B}$$

Cross multiply

$$\left[Q / (435.87 \times C) \right]^{1.853912} \times B = P_1^2 - P_2^2 - A$$

$$P_2^2 = \left[P_1^2 - A - \left(Q / (435.87 \times C) \right)^{1.853912} \times B \right]$$

Therefore;

$$P_2 = \sqrt{P_1^2 - A - \left(\left(Q / (435.87 \times C) \right)^{1.853912} \times B \right)}$$

$$P_2 = \sqrt{P_1^2 - \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right) - \left[\left(Q / (435.87 \times ((T_b/P_b)^{1.0788}) \times D^{2.6182} \times E) \right)^{1.853912} \times (G^{0.8539} \times L \times T_{avg} \times Z_{avg}) \right]}$$

Steps to calculate Outlet Pressure (P₂)

- 5) Convert the parameters to their field units;

Barg to psi (pressure)

Celsius to Rankine (temperature)

Metre to miles (length)

Metre to feet (Elevation difference)

- 6) Calculate the values of A, B and C;

$$A = \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right)$$

$$B = G^{0.8539} \times L \times T_{avg} \times Z_{avg}$$

$$C = ((T_b/P_b)^{1.0788}) \times D^{2.6182} \times E$$

- 7) Using the values of A, B and C, calculate the outlet pressure (P₂) in psi

$$P_2 = \sqrt{P_1^2 - A - \left(\left(Q / (435.87 \times C) \right)^{1.853912} \times B \right)}$$

- 8) Convert P₂ from psi to bar

7.2. Pressure Drop calculation procedure (ELEPA - AKABEL Pipeline Segment)

For determination of the required pressure drop in the ELEPA - AKABEL Pipeline Segment the Panhandle B is used with the exception of the 24" line size in which Panhandle A is used:

Panhandle B

It is used in pipelines with a diameter size greater than 24 inch, pressure values greater than 1000 psi and Reynolds number ranging from 40,000 and 40,000,000.

The Panhandle B equation is defined as;

$$Q = 737 \times (T_b/P_b)^{1.02} \times D^{2.53} \times E \times \left[(P_1^2 - P_2^2 - \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right)) / (G^{0.961} \times L \times T_{avg} \times Z_{avg}) \right]^{0.51}$$

Parameters Definition and its units

P_1 – Inlet Pressure (psia)

P_2 – Outlet Pressure (psia)

Q – Gas Flowrate (SCFD)

T_b – Base Temperature (Rankine)

P_b – Base Pressure (psia)

P_{avg} – Average Pressure (psia)

T_{avg} – Average Temperature (Rankine)

Z_{avg} – Average Gas Compressibility Factor

G – Gas specific gravity

h_2-h_1 – Elevation difference (ft)

L – Line Length (mile)

D – Inside Diameter (inch)

E – Efficiency

To find P_2 ,

Let the following parts of the equation be represented by the letters A, B and C

$$A = \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right)$$

$$B = G^{0.961} \times L \times T_{avg} \times Z_{avg}$$

$$C = \left((T_b/P_b)^{1.02} \right) \times D^{2.53} \times E$$

Therefore;

$$Q = 737 \times C \times \left[(P_1^2 - P_2^2 - A) / B \right]^{0.51}$$

Divide both sides by $737 \times C$

$$Q / (737 \times C) = \left[(P_1^2 - P_2^2 - A) / B \right]^{0.51}$$

Take the $^{0.51}\sqrt{}$ of both sides

$$\left[Q / (737 \times C) \right]^{1/0.51} = \frac{P_1^2 - P_2^2 - A}{B}$$

$$\left[Q / (737 \times C) \right]^{1.9608} = \frac{P_1^2 - P_2^2 - A}{B}$$

Cross multiply

$$\left[Q / (737 \times C) \right]^{1.9608} \times B = P_1^2 - P_2^2 - A$$

$$P_2^2 = \left[P_1^2 - A - \left(\left(Q / (737 \times C) \right)^{1.9608} \times B \right) \right]$$

Therefore;

$$P_2 = \sqrt{\left[P_1^2 - A - \left(\left(Q / (737 \times C) \right)^{1.9608} \times B \right) \right]}$$

$$P_2 = \sqrt{P_1^2 - \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right) - \left[\left(\left(Q / (737 \times \left((T_b/P_b)^{1.02} \right) \times D^{2.53} \times E \right)^{1.9608} \times \left(G^{0.961} \times L \times T_{avg} \times Z_{avg} \right) \right) \right]}$$

Steps to calculate Outlet Pressure (P₂)

5) Convert the parameters to their field units;

Barg to psi (pressure)

Celsius to Rankine (temperature)

Metre to miles (length)

Metre to feet (Elevation)

6) Calculate the values of A, B and C;

$$A = \left((0.0375 \times G \times (h_2 - h_1) \times P_{avg}^2) / T_{avg} \times Z_{avg} \right)$$

$$B = G^{0.961} \times L \times T_{avg} \times Z_{avg}$$

$$C = (T_b/P_b)^{1.02} \times D^{2.53} \times E$$

7) Using the values of A, B and C, calculate the outlet pressure (P_2) in psi

$$P_2 = \sqrt{P_1^2 - A - \left((Q / (737 \times C))^{1.9608} \times B \right)}$$

8) Convert P_2 from psi to bar

7.3. Summary and Recommendation

The pressure drop calculation is used to determine the pipeline sizing and based on these calculations, the recommended line sizes are 24 inch for BBOU – ELEPA Pipeline Segment with 28 inch and 32 inch for ELEPA – AKABEL Pipeline Segment at a FTHP of 100 barg and 80 barg respectively.

Pipeline size (inch) BBOU - ELEPA	Pipeline size (inch) ELEPA - AKABEL	Inlet Pressure (barg) at BBOU	Inlet Pressure (barg) at ELEPA	Delivery Pressure (barg) at AKABEL
24	28	100	97.77	63.01
24	32	80	78.00	62.14